

Assignment Title: Assignment 2

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Date: _____

Question #1)

$$2. \int 3\sqrt{4+2x} \, dx$$

$$\int 3\sqrt{4+2x} \, dx \rightarrow \int 3(4+2x)^{\frac{1}{2}} = \frac{3}{\frac{3}{2}} (4+2x)^{\frac{3}{2}} \cdot \frac{2}{3} + c$$

$$\boxed{(4+2x)^{\frac{3}{2}} + c} \quad \neq$$

$$4. \int 4x \tan(x^2) \, dx$$

$$\text{Let } u = x^2 \\ du = 2x \, dx$$

$$2 \int \tan u \, du = \boxed{-2 \ln(\cos x^2) + c}$$

$$6. \int \frac{1}{9+4x^2} = \frac{1}{4} \left[\frac{1}{9/4 + x^2} \right] = \frac{1}{4} \left[\frac{1}{x^2 + (\frac{3}{2})^2} \right]$$

$$\therefore \int \frac{1}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$$

$$\frac{1}{4} \left[\frac{2}{3} \tan^{-1} \frac{2x}{3} \right] + c = \boxed{\frac{1}{6} \tan^{-1} \frac{2x}{3} + c}$$

$$8. \int \frac{\sec(\ln x) \tan(\ln x)}{x} \, dx$$

$$\text{Let } u = \ln x \quad du = \frac{1}{x} \, dx$$

$$\int \sec u \tan u \rightarrow \boxed{\sec(u) + c} \\ \boxed{\sec(\ln x) + c}$$

10. $\frac{1}{2} \int \frac{2x}{\sqrt{1-x^2}} dx$

Let $u = x^2$ $du = 2x dx$

$\int \frac{du}{\sqrt{1-u}} = \sin^{-1}(u) + c$

$\frac{1}{2} \int \frac{du}{\sqrt{1-u^2}} \rightarrow \boxed{\frac{1}{2} \sin^{-1}(x^2) + c}$

12. $\int \frac{\cos x}{\sin x \sqrt{\sin^2 x + 1}}$

Let $u = \sin x$ $du = \cos x dx$

$\int \frac{du}{u \sqrt{u^2+1}}$

So now let $t = \sqrt{u^2+1}$ $t^2 = u^2+1$

$2t dt = 2u du$

$\frac{du}{u} = \frac{dt}{t^2} = \int \frac{dt}{t^2} = -\frac{1}{t} + c$

$-\frac{1}{\sqrt{u^2+1}} + c = \boxed{-\frac{1}{\sqrt{\sin^2 x + 1}} + c}$

14. $\int \frac{e^{\tan^{-1}x}}{1+x^2} dx$ let $\tan^{-1}(x) = u$ $du = \frac{1}{1+x^2}$

$\int \frac{e^u}{1+x^2} du = e^u + c$ $e^{\tan^{-1}x} + c$

16. $\int (x+1) \cot(x^2+2x) dx$

let $u = x^2 + 2x$ $du = 2x + 2 \rightarrow 2(x+1) du$

$= \frac{1}{2} \int (\cot u) du \rightarrow \frac{1}{2} \ln(\sin u) + c = \boxed{\frac{1}{2} \ln \sin(x^2+2x) + c}$

18. $\int \frac{dx}{x(\ln x^2)}$ let $u = \ln x$ $du = \frac{dx}{x}$

$\int u^{-2} du \rightarrow -u^{-1} + c$

$-\frac{1}{\ln x} + c$

$\int \frac{dx}{\sqrt{x^2-4}} = \int \frac{dx}{\sqrt{x^2-a^2}} = \ln(x + \sqrt{x^2-a^2}) + c$

$\ln(x + \sqrt{x^2-4}) + c$

24. $\int \frac{\cos(\ln x)}{x} dx$ let $\ln x = u$ $du = \frac{1}{x} dx$

$$\int \frac{\cos(\ln x)}{x} dx = \int \cos u \Rightarrow \sin u + c$$

$$\boxed{\sin(\ln x) + c}$$

26. $\int \frac{\sinh(x^{-1/2})}{x^{3/2}} dx$

$$\int \sinh(u) \cdot 2(du) = 2 \int \sinh(u)$$

$$-2 \cosh u + c \Rightarrow \boxed{-2 \cosh(x^{-1/2}) + c}$$

28. $\int \frac{e^x}{\sqrt{4-e^{2x}}} dx$ let $e^x = u$ $du = e^x dx$ " $e^{2x} = u^2$ "

$$\Rightarrow \int \frac{1}{\sqrt{4-u^2}} \Rightarrow \sin^{-1}\left(\frac{u}{2}\right) + c$$

$$\frac{1}{\sqrt{4-u^2}}$$

$$\int \frac{1}{\sqrt{4-u^2}} = \sin^{-1}\left(\frac{u}{2}\right) + c \Rightarrow \boxed{\sin^{-1}\left(\frac{e^x}{2}\right) + c}$$

3a. $\int 2^{\pi x}$

$$2^{\pi x} = e^{\pi x \ln 2}$$

$$2^{\pi x} = \int e^{\pi \ln 2 u} du$$

$$u = \pi \ln 2$$

$$\int e^{u x} = \frac{1}{u} e^{u x} + c$$

$$\frac{1}{\pi \ln 2} e^{\pi x \ln 2} + c$$

$$e^{\pi x \ln 2} = 2^{\pi x} \rightarrow$$

$$\boxed{\frac{2^{\pi x}}{\pi \ln 2} + c}$$

Question #2

2. $\int x e^{3x}$

$$(x)(e^{3x})(1/3) - 1 \cdot \int \frac{e^x}{3} = \frac{x(e^{3x})}{3} - \frac{1}{9} e^{3x} + c$$

$$\boxed{\frac{e^{3x}(3x-1)}{9} + c}$$

4. $\int x^2 \cdot e^{-2x} dx$

$$x^2 \cdot \frac{e^{-2x}}{2} - \int \left((2x) \cdot \left(\frac{e^{-2x}}{2} \right) \right) = (x) \left(\frac{e^{-2x}}{-2} \right) - \int 1 \cdot \frac{e^{-2x}}{2}$$

$$-\frac{x^2 e^{-2x}}{2} + \int x e^{-2x}$$

$$= \boxed{\frac{-e^{-2x}(2x^2 + 2x + 1)}{4} + c}$$

$$= -\frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4}$$

6. $\int x \cdot \cos 2x$

$$x(\sin 2x)\left(\frac{1}{2}\right) - \int 1 \cdot \sin 2x \cdot \frac{1}{2}$$

$$\boxed{\frac{1}{2}x(\sin 2x) + \frac{1}{4} \cos 2x + c}$$

8. $\int x^2 \cdot \sin x$

$$x^2(-\cos x) - \int 2x \cdot -\cos x \Rightarrow -x^2 \cos x + 2 \int x \cdot \cos x$$

$$(-x^2)(\cos x) + 2x \sin x + 2 \cos x + c$$

$$\boxed{-x^2 \cos x + 2x \sin x + 2 \cos x + c}$$

10. $\int \sqrt{x} \ln x$

$$(\ln x) \left(\frac{2}{3} x^{3/2} \right) - \int \frac{1}{x} \cdot \frac{2}{3} x^{3/2}$$

$$\frac{2}{3} x^{3/2} \ln x - \frac{2}{3} x^{1/2} \Rightarrow \frac{2}{3} x^{3/2} \ln x - \frac{4}{9} x^{3/2} + c$$

$$\boxed{\frac{2}{3} x^{3/2} \left(\ln x - \frac{2}{3} \right) + c}$$

$$12. \int \frac{\ln x}{\sqrt{x}} \Rightarrow \int (\ln x) (x^{-1/2}) \Rightarrow (\ln x) (2\sqrt{x}) - \int \frac{1}{x} (2\sqrt{x})$$

$$\Rightarrow (2\sqrt{x})(\ln x) - 4\sqrt{x} + c$$

$$\boxed{2\sqrt{x} (\ln x - 2) + c}$$

$$14. \int \ln(x^2+4) dx$$

$$\ln(x^2+4) \times x - \int \frac{1}{x^2+4} \cdot 2x \cdot x$$

$$x \cdot \ln(x^2+4) - 2 \int \frac{x^2}{x^2+4}$$

$$\frac{x^2}{x^2+4} = 1 - \frac{4}{x^2+4} = x \cdot \ln(x^2+4) - 2x + 4 \tan^{-1}\left(\frac{x}{2}\right) + c$$

$$\boxed{x \ln(x^2+4) - 2x + 4 \tan^{-1}\left(\frac{x}{2}\right) + c}$$

$$16. \int \cos^{-1}(2x)$$

$$\cos^{-1}(2x) \cdot x - \int \frac{1}{\sqrt{1-4x^2}} \times x$$

$$x(\cos^{-1}(2x)) + 2 \frac{x}{\sqrt{1-4x^2}}$$

$$\text{let } t = 1 - 4x^2$$

$$dt = -8x dx$$

$$\int -\frac{1}{2} \sqrt{1-4x^2}$$

$$\boxed{x \cos^{-1}(2x) - \frac{1}{2} \sqrt{1-4x^2} + c}$$

18. $\int x \cdot \tan^{-1}(x)$

$$\tan^{-1}(x) \cdot \left(\frac{x^2}{2}\right) - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2}$$

$$\frac{x^2}{2} \tan^{-1}(x) - \frac{1}{2} \int \frac{x^2}{1+x^2}$$

$$\frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$$

$$\frac{x^2}{2} \tan^{-1}(x) - \frac{1}{2} (x - \tan^{-1}(x)) + c$$

$$\boxed{\frac{(1+x^2)}{2} \tan^{-1}(x) - \frac{x}{2} + c}$$

20. $\int e^{3x} \cos 2x$

$$(e^{3x}) (\sin 2x) \left(\frac{1}{2}\right) - \int e^{3x} \cdot 3 \sin x \left(\frac{1}{2}\right)$$

$$\frac{1}{2} (e^{3x}) (\sin 2x) - \frac{1}{2} \int (3 e^{3x}) (\sin 2x)$$

$$\frac{1}{2} (e^{3x}) (\sin 2x) - \frac{3}{2} \int e^{3x} (\sin 2x)$$

$$\frac{e^{3x} (3 \cos 2x + 2 \sin 2x)}{13} + c$$

22. $\int \cos(\ln x) dx$ Let $t = \ln x$ $dt = \frac{dx}{x}$ $dx = e^t dt$

$$\int e^t \cdot \cos t = (e^t)(\sin t) - \int e^t \sin t dt$$

$$\boxed{\frac{x [\sin(\ln x) + \cos x \ln x] + c}{2}}$$

24. $\int x \cdot \tan^2 x$

$$x \sec^2 x dx - x$$

$$(x \cdot \sec^2 x) = (x)(\tan x) - \int 1(\tan x) = x \tan x + \ln |\cos x|$$

$$\boxed{\frac{x \tan x + \ln(\cos x) - \frac{x^2}{2} + c}{2}}$$

26. $\int \frac{x e^x}{(x+1)^2} \Rightarrow \frac{d}{dx} \left(\frac{e^x}{x+1} \right) = \frac{(x+1)(e^x) - (e^x)(1)}{(x+1)^2}$

$$\Rightarrow \frac{x e^x}{(x+1)^2}$$

$$\boxed{\int \frac{x e^x}{(x+1)^2} = \frac{e^x}{x+1} + c}$$

28. $\int_0^1 x e^{-5x}$

$$(x)(e^{-5x})(-1/5) - \int 1 \cdot e^{-5x}(-1/5)$$

$$-1/5 (x)(e^{-5x}) + \frac{1}{5} \int e^{-5x} + c \Rightarrow \left| \frac{x e^{-5x}}{5} - \frac{e^{-5x}}{25} \right|_0^1$$

$$\boxed{\frac{1 - 6e^{-5}}{25}}$$

30. $\int_{\sqrt{e}}^e \frac{\ln x}{x^2} dx$ let $t = \ln x$ $dx = e^t dt$
 $x = e^t$

$$(t)(e^{-t}) - \int (1)(e^{-t}) \Rightarrow -te^{-t} + \int e^{-t}$$

$$-te^{-t} - e^{-t} + c = -(t+1)e^{-t} + c$$

When $x = \sqrt{e} \rightarrow t = 1/2$

When $x = e \rightarrow t = 1$

$$\boxed{\frac{3}{2e^{1/2}} - \frac{2}{e}}$$

ues # 3

2. $\sqrt{1-4x^2} \rightarrow \sqrt{(1)^2 - 2x^2}$

$2x = \sin u \rightarrow x = \frac{1}{2} \sin u$ $dx = \frac{1}{2} \cos u$ then

$\sqrt{1-4x^2} = \cos u$

$1-4x^2 = \sin$

$\sqrt{1-4x^2} = \sqrt{1-\sin^2 u} = \cos u$

So,

$$\sqrt{1-4x^2} = \int \cos u \cdot \frac{1}{2} \cos u = \frac{1}{2} \cos^2 u$$

$$\boxed{\frac{1}{4} \sin^{-1}(2x) + \frac{1}{2} x \sqrt{1-4x^2} + c}$$

Date: _____

4. $\int \frac{dx}{x^2 \sqrt{9-x^2}}$ Let $x = 3 \sin \theta$ $dx = 3 \cos \theta d\theta$
 $\sqrt{9-x^2} = 3 \cos \theta$

$$\frac{3 \cancel{\cos \theta}}{9 \sin^2 \theta \cdot 3 \cancel{\cos \theta}} = \frac{1}{9} \int \operatorname{cosec}^2 \theta = -\frac{1}{9} \cot \theta + c$$

$$\cot \theta = \frac{(3)(\sqrt{9-x^2})}{3(x)} = \cot \theta = \frac{\sqrt{9-x^2}}{x}$$

$$\boxed{-\frac{\sqrt{9-x^2}}{9x} + c}$$

6. $\int \frac{x^2}{\sqrt{5+x^2}}$ Let $x^2 = (x^2+5) - 5$

$$\sqrt{5+x^2} = -5 \int \frac{dx}{5+x^2}$$

$$\therefore \sqrt{a^2+x^2} = \frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \ln(x + \sqrt{a^2+x^2})$$

So,

$$\boxed{\frac{x}{2} \sqrt{5+x^2} - \frac{5}{2} \ln(x + \sqrt{5+x^2}) + c}$$

8. $\int \frac{dx}{x^2 \sqrt{x^2-16}}$ $x = 4 \sec u$ $dx = 4 \sec u \tan u$ $x^2-16 = 4 \tan u$

$$\frac{4 \sec u \cdot \tan u}{16 \sec^2 u \cdot 4 \tan u} = \frac{1}{16} \int \cos u = \frac{1}{16} \sin u + c$$

$$\sin u = \frac{\sqrt{x^2-16}}{x} = \boxed{\frac{1}{6} \frac{\sqrt{x^2-16}}{x} + c}$$

10. $\int x^3 \sqrt{5-x^2}$ Let $u = 5-x$ $du = -2x$ $x du = -\frac{du}{2}$
 $x^2 = 5-u$

$$x^3 \sqrt{5-x^2} = \int x^2 \cdot x \sqrt{u} = \int (5-u) \sqrt{u} \left(-\frac{1}{2} du\right) =$$

$$-\frac{1}{2} \int 5u^{-1/2} - u^{3/2}$$

$$-\frac{1}{2} \left[5 \cdot \frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} + c \right] = \frac{5}{3} u^{3/2} + \frac{1}{5} u^{5/2} + c$$

$$\boxed{-\frac{5}{3} (5-x^2)^{3/2} + \frac{1}{5} (5-x^2)^{5/2} + c}$$

12. $\int \frac{1+t^2}{t}$ Let $u = \sqrt{1+t^2}$ $u^2 = 1+t^2$
 $2u du = 2t dt$
 $dt = \frac{u}{t}$

$$\frac{u \cdot u}{t \cdot t} = \frac{u^2}{t^2} = \frac{u^2}{u^2-1}$$

$$\frac{u^2}{u^2-1} = 1 + \frac{1}{u^2-1}$$

$$\int 1 + \int \frac{1}{u^2-1} = u + \frac{1}{2} \ln \left| \frac{u-1}{u+1} \right| + c$$

$$\boxed{\sqrt{1+t^2} + \frac{1}{2} \ln \left| \frac{\sqrt{t^2+1}-1}{\sqrt{t^2+1}+1} \right| + c}$$

Let $x = 5 \tan u$ $dx = 5 \sec^2 u$ $\sqrt{x^2 + 25}$

14. $\int \frac{dx}{x^2 \sqrt{x^2 + 25}}$

$$\int \frac{5 \sec^2 u}{25 \tan^2 u \cdot 5 \sec u} \rightarrow \frac{1}{25} \int \frac{\sec u}{\tan^2 u}$$

$$\frac{\sec u}{\tan^2 u} = \frac{1/\cos u}{\sin^2 u / \cos^2 u} \cdot \frac{\cos u}{\sin^2 u} = \cot u \operatorname{cosec} u$$

$$\frac{1}{25} \int \cot u \cdot \operatorname{cosec} u \Rightarrow -\frac{1}{25} \operatorname{cosec}^2 + c$$

$$\boxed{-\frac{1}{25} \frac{\sqrt{x^2 + 25}}{x} + c}$$

16. $\int \frac{dx}{1 + 2x^2 + x^4}$ $\int \frac{dx}{(1+x^2)^2}$ $x = \tan u$ $dx = \sec^2 u$
 $1 + x^2 = \sec^2 u$
 $= \cos^2 u$

$$\int \cos^2 u \Rightarrow \frac{1}{2} u + \frac{1}{4} \sin 2u + c$$

$$\left\{ \int u = \tan^{-1}, \sin 2u = \frac{2x}{1+x^2} \right\}$$

$$\boxed{= \frac{1}{2} \tan^{-1}(x) + \frac{x}{2(1+x^2)} + c}$$

18. $\int \frac{3x^3}{\sqrt{x^2-25}}$ Date: let $x = 5 \sec \theta$ $dx = 5 \sec \theta \tan \theta$
 $\sqrt{x^2-25} = 5 \tan \theta$

$$\frac{3(125 \sec^3 \theta)}{5 \tan \theta} \cdot 5 \sec \theta \tan \theta = 375 \int \sec^4 \theta$$

$$375 \int 1 + \tan^2 \theta \cdot \sec^2 \theta \quad \text{let } u = \tan \theta \quad du = \sec^2 \theta$$

$$375 \int (1 + u^2) = 375 \left[u + \frac{u^3}{3} \right] + c$$

$$\left\{ \begin{aligned} u &= \tan \theta = \sqrt{x^2-25} \end{aligned} \right\}$$

$$375 \left[\frac{\sqrt{x^2-25}}{5} + \frac{(x^2-25)^{3/2}}{375} + c \right]$$

$$\boxed{(x^2-25)^{3/2} + 75 \sqrt{x^2-25} + c}$$

20. $\int \frac{\cos \theta}{\sqrt{2-\sin^2 \theta}}$ let $u = \sin \theta$ $du = \cos \theta$
 $\frac{du}{\sqrt{a^2-u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + c$

$$\int \frac{du}{\sqrt{2-u^2}}$$

$$\boxed{\sin^{-1} \left(\frac{\sin \theta}{\sqrt{2}} \right) + c}$$

$$22. \int_0^{1/2} \frac{dx}{(1-x^2)^2} \quad \text{Let } x = \sin t \quad dx = \cos t \, dt$$

$$1-x^2 = \cos^2 t \quad x=0, t=0$$

$$x=1/2, t=\pi/6$$

$$\int \sec^3 t = \frac{1}{2} (\sec t \tan t + \ln |\sec t + \tan t| + C)$$

$$t=0 \quad \sec 0 = 1 \quad \tan 0 = 0 \quad \ln 1 = 0$$

$$\frac{1}{2} \left(\frac{2}{3} + \ln 3 \right) - 0 \Rightarrow \boxed{\frac{1}{3} + \frac{1}{4} \ln 3}$$

$$24. \int_{\sqrt{2}}^2 \frac{\sqrt{2x^2-4}}{x} \quad 2x^2-4 = \sqrt{2} \sqrt{x^2-2} \quad x = \sqrt{2} \sec u$$

$$\sqrt{x^2-2} = \sqrt{2} \tan u$$

$$x = \sqrt{2} \sec u = 1, u=0$$

$$x = 2 \quad \sec u = \frac{2}{\sqrt{2}} = \sqrt{2} \quad u = \frac{\pi}{4}$$

$$\int_0^{\pi/4} 2 \tan^2 u = 2 \int (\sec^2 u - 1) = 2 \left(\tan u - u \right) \Big|_0^{\pi/4}$$

$$\left(2 \tan\left(\frac{\pi}{4}\right) - \frac{\pi}{4} \right) - 0$$

$$2 \left(1 - \frac{\pi}{4} \right) \Rightarrow \boxed{2 - \frac{\pi}{2}}$$

Ques #4

Date:

38.

$$\int \frac{dx}{\sqrt{2x-x^2}}$$

$$2x-x^2 = 1-(x-1)^2$$

~~$$= \sin^{-1}(x-1) + c$$~~

$$\frac{d}{dx} \sin^{-1}(x-1) = \frac{1}{\sqrt{1-(x-1)^2}}$$

$$\boxed{\frac{dx}{1-(x-1)^2} = \sin^{-1}(x-1) + c}$$

40.

$$\int \frac{dx}{16x^2+16x+5}$$

$$16x^2+16x+5 = 16\left(x+\frac{1}{2}\right)^2+1$$

$$u = 4\left(x+\frac{1}{2}\right) = 4x+2$$

$$du = dx = \frac{du}{4}$$

$$\frac{1}{4} \left(\frac{du}{u^2+1} \right) = \frac{1}{4} \tan^{-1}(u) + c = \frac{1}{4} \tan^{-1}(4x+2)$$

$$\boxed{\frac{1}{4} \tan^{-1}(4x+2) + c}$$

42.

$$\int \frac{x}{x^2+2x+2}$$

$$x = (x+1) - 1$$

$$\int \frac{x+1}{x^2+2x+2} - \int \frac{1}{x^2+2x+2}$$

$$u = x^2+2x+2 \quad du = 2(x+1) = \frac{1}{2} \ln u$$

$$\Rightarrow x^2+2x+2 = (x+1)^2 + 1$$

$$\int \frac{1}{(x^2+1)^2+1} = \tan^{-1}(x+1)$$

$$\boxed{\frac{1}{2} \ln(x^2+2x+2) - \tan^{-1}(x+1) + c}$$

let $u = e^{2x}$ $du = e^{2x} \cdot 2 dx$

44. $\int \frac{e^x}{1+e^x+e^{2x}}$

$\int \frac{du}{\sqrt{u^2+u+1}} \quad \left\{ u^2+u+1 = \left(u+\frac{1}{2}\right)^2 + \frac{3}{4} \right\}$

$\int \frac{du}{\sqrt{u^2+a^2+b^2}} = \ln(u+a+\sqrt{(u+a)^2+b^2}) + c$

$\ln\left(e^x + \frac{1}{2} + \sqrt{1+e^x+e^{2x}}\right) + c$

46. $\int \frac{2x+3}{4x^2+4x+5}$ let $t = x + \frac{1}{2}$ then $4x^2+4x+5 = 4(t^2+1)$
 $2x+3 = 2(t-\frac{1}{2})+3 = 2t+2 = 2(t+1)$

$\frac{2(t+1)}{4(t^2+1)} = \frac{t}{2(t^2+1)} + \frac{1}{2(t^2+1)}$

$\int \frac{t}{2(t^2+1)} = \frac{1}{4} \ln(t^2+1) = \int \frac{1}{2(t^2+1)} = \frac{1}{2} \tan^{-1}(t)$

$\frac{1}{4} \ln\left(x + \frac{1}{2}\right)^2 + \frac{1}{2} \tan^{-1}\left(x + \frac{1}{2}\right) + c$

48. $\int_0^4 \sqrt{x(4-x)}$ let $x = 4\sin^2\theta$ $dx = 8\sin\theta\cos\theta$, $\sqrt{x(4-x)} = 4\sin\theta\cos\theta$

So, $\int_0^{\pi/2} (4\sin\theta\cos\theta)(8\sin\theta\cos\theta) = 32 \int_0^{\pi/2} \sin^2\theta\cos^2\theta$

$\sin^2\cos^2\theta = \frac{1}{8}(1-\cos 4\theta) \Rightarrow 32 \cdot \frac{1}{8} \int_0^{\pi/2} 1 - \cos 4\theta = 4\left(\frac{\pi}{2} - \frac{1}{4}\sin 4\theta\right) \Big|_0^{\pi/2}$

$\Rightarrow 4 \cdot \frac{\pi}{2} \rightarrow \boxed{2\pi}$